

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech****Branch : Electronics and Computer Engineering / Electronics and Computer Science Engineering****Semester : VI Subject Code & Name: BTECPC601 Internet of Things****Max Marks: 60****Date:18/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	IoT is based on _____ technology		1
	a. Hardware b. Software c. Both a & b d. None		
2	Which of the following are the features of cloud computing?		1
	a. Security b. Availability c. Large Network Access d. All of the above		
3	Which of the following are the applications of wireless sensor network (WSN)?		1
	a. Military b. Health monitoring c. Industries d. All of the above		
4	Which of the following is not a variable type in c programming?		1
	a. Int b. Float c. While loop d. All of the above		
5	Which keyword is used for function in Python language?		1
	a. Function b. Fun c. def d. Define		
6	The Raspberry Pi 3 Model used _____ bit processor		1
	a.32 b. 64 c. 128 d. None_		
7	Which of the following is not core data type in Python programming?		1
	a. List b. Tuple c. Dictionary d. Class		
8	Which of the following is not an arithmetic operation?		1
	a. a * = 10; b. a / = 10 c. a != 10 d. a % = 10		

9	Which instruction set architecture is used in Raspberry Pi?					1
	a. MSP	b. X86	c. AVR	d. ARM		
10	MQTT protocol is mainly used for.....					1
	a. M2M communication	b. Device communication	c. Internet communication	d. Wireless communication		
11	XMPP uses _____ architecture.					1
	a. Decentralized client-server	b. Centralized client-server	c. Message	d. Public / Subscriber		
12	How many GPIO pins are present in NodeMCU (ESP8266)					1
	a. 17	b. 15	c. 10	d. 20		
Q. 2	Solve the following.					12
A)	What is an API? Explain REST and Web socket IoT communication APIs					6
B)	Explain the Cloud Computing architecture and give its applications.					6
Q.3	Solve the following.					12
A)	Explain the NodeMCU 8266 Wi-Fi module with suitable pin diagram					6
B)	Write a note on debugging and optimization of C programs					6
Q. 4	Solve Any Two of the following.					12
A)	Discuss the different data types available in Python with suitable examples					6
B)	Explain basic architecture of Raspberry Pi and give its popular applications.					6
C)	Write short note on basic of Linux commands					6
Q.5	Solve Any Two of the following.					12
A)	Explain the interfacing of R Pi 3 with twitter and whatsapp					6
B)	Explain how data can publish and subscribe from web using R pi3					6
C)	Give the steps for configuring NodeMCU to connecting to a server					6

Q. 6	Solve Any Two of the following.		12
A)	Draw and explain the architecture of Data Distribution Service (DDS) protocol used in IoT		6
B)	Compare in between Message Queue Telemetry Transport (MQTT) protocol and Advanced Message Queuing Protocol (AMQP).		6
C)	Write short note on Bluetooth Low Energy (BLE) protocol		6
	*** End ***		